

D 130229

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Name.....

Reg. No.....

**FIFTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2025**

Mathematics

MTS 5B 08—LINEAR PROGRAMMING

(2020 Syllabus)

Time : Two Hours

Maximum : 60 Marks

Section A (Short Answer Type)*All questions can be answered.**Each question carries 2 marks.**(Ceiling 20 marks)*1. The equation of the line segment joining $(2, 1)$ and $(5, 7)$ is _____.2. Define a hyperplane in $\mathbb{R}^2$.

3. Convert the linear programming problems given below to canonical form :

Maximize $f(x, y) = -x - 2y$ subject to
 $2x - y \geq 10$ $3y - x \leq 8$, $x \geq 0, y \geq 0$.

4. Draw and shade a convex subset of $\mathbb{R}^2$ having no extreme points.

5. Consider the canonical maximum tableau below :

x	y	-1	
3	1	5	$= -t_1$
-7	1	8	$= -t_2$
4	9	13	$= f$

State the canonical maximization linear programming problem represented by the tableau above.

Turn over

6. Write the basic solution of the linear programming problem given below :

Maximize $f(x, y) = 200x + 150y$ subject to
 $x + 2y \leq 11$, $2x + 2y \leq 30$, $2x + 2y \leq 25$ $x \geq 0, y \geq 0$.

7. Write the canonical slack minimization linear programming problem.
 8. State Von-Neumann Minimax Theorem.
 9. What is complementary slackness of a dual canonical linear programming problem ?
 10. What is balanced transportation problem ?
 11. What is the pure strategy of a matrix game ?
 12. What is a basic feasible solution of a transportation problem ?

Section B (Paragraph/Problem Type)

*All questions can be answered.
 Each question carries 5 marks.
 (Ceiling 30 marks)*

13. Show that the linear programming problem :

Maximize $f(x, y) = 3x + 2y$ subject to
 $2x - y \leq -1$, $x - 2y \geq 0$, $x \geq 0, y \geq 0$.

is infeasible.

14. Solve the linear programming problem graphically :

Maximize $f(x, y) = 5x + 2y$ subject to
 $x + 3y \leq 14$, $2x + y \leq 8$, $x \geq 0, y \geq 0$.

15. Apply the simplex algorithm to the maximum tableau given below :

x_1	x_2	-1	
2	1	8	$= -t_1$
1	2	10	$= -t_2$
30	50	10	$= f$

16. Solve the unconstrained problem :

Maximize $f(x, y) = x + 3y$ subject to
 $x + 2y \leq 10, \quad -3x - y \leq -15.$

17. A tableau of the balanced assignment problem is given below :

1	0	0
0	1	0
1	0	1

Find a permutation set of zeros.

18. Solve the assignment problem given below :

38	21	34
41	14	36
28	20	25

19. What is a two-person zero-sum matrix game ?

Section C (Essay Type)

Answer any **one** of the following question.

The question carries 10 marks.

20. Solve the transportation problem by using Vogel Advanced-Start Method :

	M ₁	M ₂	M ₃	
W ₁	2	1	2	40
W ₂	9	4	7	60
W ₃	1	2	9	10
	40	50	20	110

Turn over

21. A company wishes to assign five of its workers to five different jobs. The rating of each worker with respect to each job on a scale of 0 to 10 is given by the following table :

	J_1	J_2	J_3	J_4	J_5
W_1	5	4	2	8	5
W_2	7	6	4	6	9
W_3	5	5	3	3	2
W_4	4	3	5	5	4
W_5	3	6	4	10	2

If the company wishes to maximize the total rating of the assignment, find the optimal assignment plan and the corresponding maximum total rating.

(1 × 10 = 10 marks)